

## Healthcare Resource Availability: Analyzing Global Disparities

### ADAN8880 - Directed Practicum in Analytics

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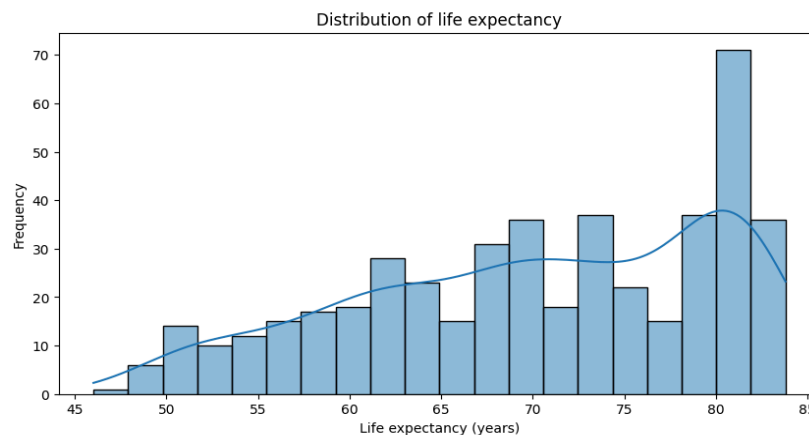
#### Dataset partition strategy

For this week, we partitioned the dataset into a training set (70%) from 2002-2015, a validation set (15%) from 2016-2018, and a test set (15%) from 2019-2021 to reflect the time-series nature of the data. This chronological split helps our models to learn from enough historical trends and generalize to future periods. This approach is important for identifying long-term patterns in healthcare disparities across income groups and also ensure the model remains robust to real-world changes. The validation set helps us to fine-tune models, and the test set allows us to evaluate performance under unforeseen global disruptions, such as the COVID-19 pandemic. We expect the model to capture systemic healthcare inequalities across different economic groups. By structuring the data this way, we ensure our analysis remains relevant for policymaking and global health initiatives aimed at addressing systemic inequalities.

#### EDA Analysis

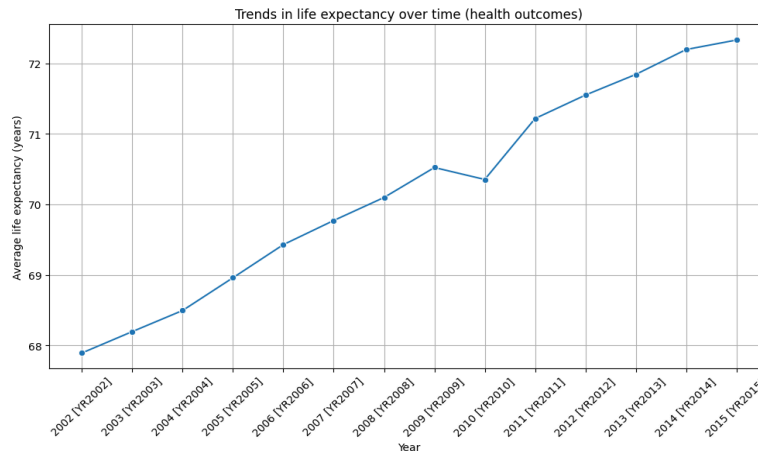
For our EDA analysis, we included visualizing distributions, identifying trends, analyzing relationships between key healthcare indicators, and examining external influences like the 2008 economic crisis.

1. We plotted a histogram to observe how life expectancy is distributed across countries



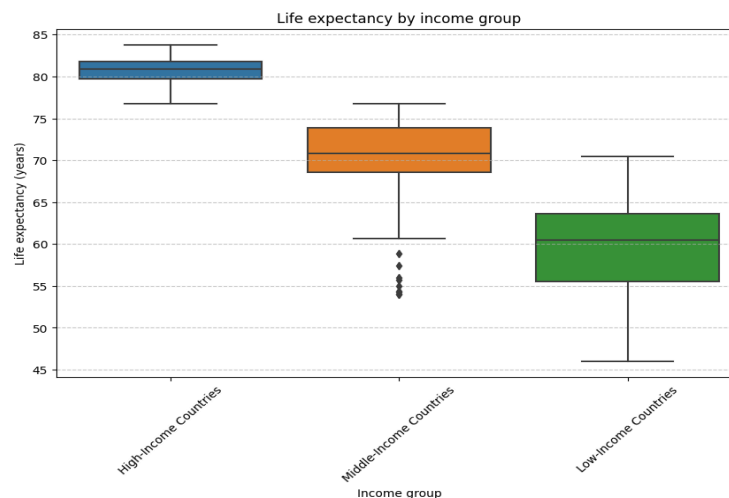
- The distribution of life expectancy shows a right-skewed pattern. Life expectancy is not evenly distributed globally, with low-income countries significantly behind and struggle with low longevity.
- The distribution shows two clusters, one around 60–70 years (likely middle and low-income countries) and another near 80 years (high-income countries).

2. We analyzed the year-over-year changes in average life expectancy from training data.



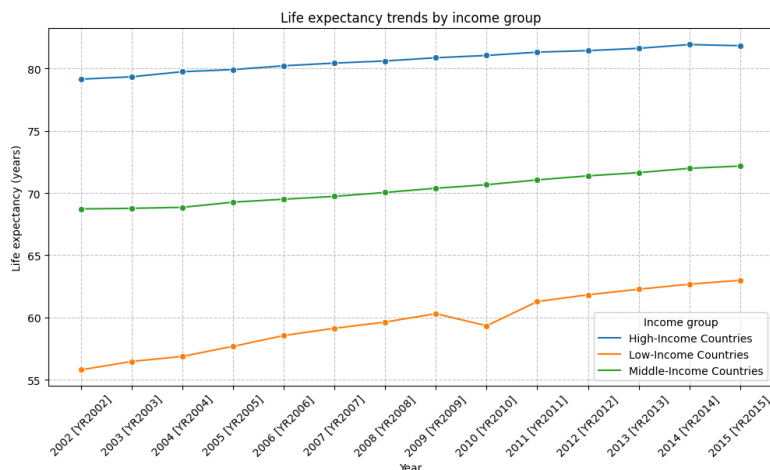
- Life expectancy rose from approximately 68 years in 2002 to around 72 years in 2015. However, there was a noticeable decline around 2010, which may reflect economic disruptions following the 2008 financial crisis, but it rebounded quickly.
- There is a steeper increase after 2010, which suggests that post-crisis healthcare policies may have played a role in improving longevity.

3. We used a box plot to compare life expectancy across high, middle, and low-income countries.



- High-income countries have the highest median life expectancy around 81 years with low variability, which indicates they have consistent healthcare access and quality.
- Middle-income countries have a median of about 71 years and show more variation, with some countries having life expectancies closer to low-income countries.
- Low-income countries have the lowest median at approximately 60 years, with the widest range.
- Low and middle-income countries show larger spreads, with many outliers observed in middle-income countries, meaning life expectancy varies widely within these groups, which reflects significant disparities in healthcare access and development.

4. We plotted life expectancy trends separately for high, middle, and low-income countries to examine differences in growth.



This graph shows life expectancy trends (2002–2015) for high-, middle-, and low-income countries, highlighting disparities and growth patterns. **High-income countries (blue)** have the highest life expectancy, starting above 80 years with steady growth and low variability, reflecting strong healthcare systems. **Middle-income countries (green)** start around 70 years and grow to 73 years by 2015, with slightly higher variability due to mixed healthcare access. **Low-income countries (orange)** begin at 55 years and grow to 60 years by 2015, showing slower progress and a dip around 2010, likely due to global challenges like economic instability. This graph shows life expectancy trends (2002–2015) for high-, middle-, and low-income countries. High-income countries start above 80 years with steady growth, middle-income countries grow from 70 to 73 years, and low-income countries rise from 55 to 60 years, with slower progress and a dip around 2010 due to global challenges.

5. We conducted an outlier analysis by identifying the top 5 and bottom 5 countries based on average life expectancy across years.

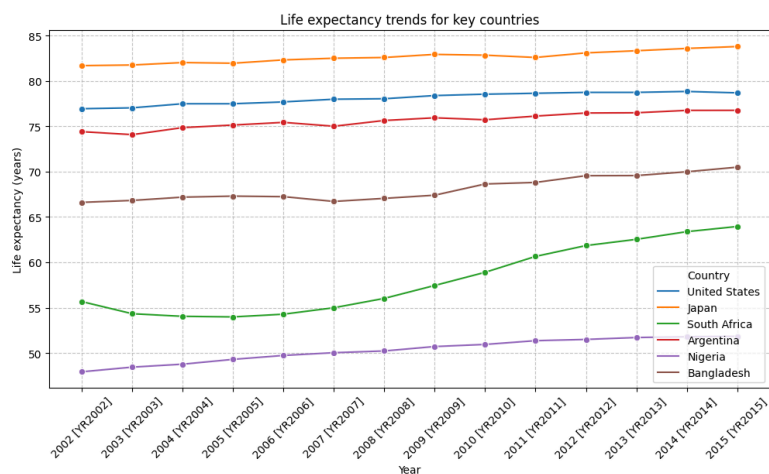
Top 5 countries with highest life expectancy

| Country name | Average Life Expectancy |
|--------------|-------------------------|
| Japan        | 82.644843               |
| Switzerland  | 81.931882               |
| Italy        | 81.600174               |
| Australia    | 81.376132               |
| Sweden       | 81.175610               |

## Bottom 5 countries with lowest life expectancy

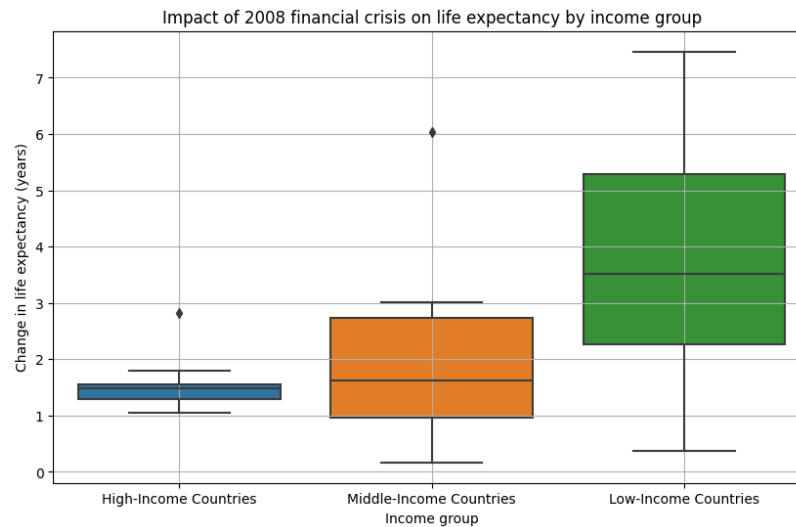
| Country name | Average Life Expectancy |
|--------------|-------------------------|
| Nigeria      | 50.305071               |
| Mozambique   | 53.740857               |
| Uganda       | 55.824929               |
| Ethiopia     | 57.870357               |
| South Africa | 58.001286               |

- The 40+ year gap between the highest and lowest life expectancy countries highlights deep global health inequities. High-life-expectancy countries remain stable around 83 years, and low-life-expectancy countries fall below 60 years and show much greater variation. This contrast underscores systemic disparities, where certain countries consistently face challenges in healthcare access that significantly lower life expectancy.
6. We selected representative countries from high, middle, and low-income groups to analyze life expectancy trends across different economic levels.

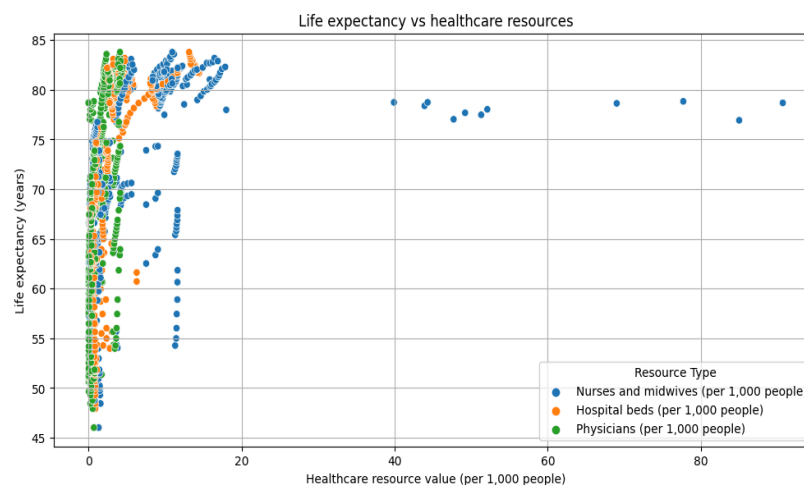


- High-income: Japan, United States ; Middle-income: Argentina, Bangladesh ; Low-income: Nigeria, South Africa
- These countries represent their income groups: High-income countries (e.g., Japan, U.S.) show stable trends, middle-income countries display steady growth, and low-income countries start lower but show improvements with greater variability.

7. We analyzed how life expectancy changed before and after the 2008 crisis across different income groups.

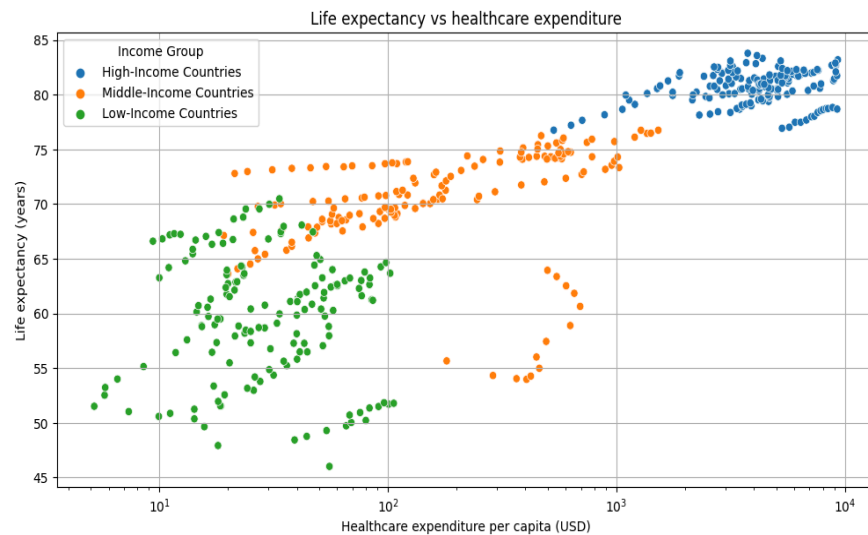


- The 2008 financial crisis did not harm life expectancy; all income groups saw increases. Low-income countries had the largest gains and widest impacts, middle-income countries showed moderate, variable gains, and high-income countries had smaller but stable increases. This highlights the role of healthcare improvements and long-term trends over economic instability.
8. We examined how healthcare resources (physicians, hospital beds, nurses) correlate with life expectancy.

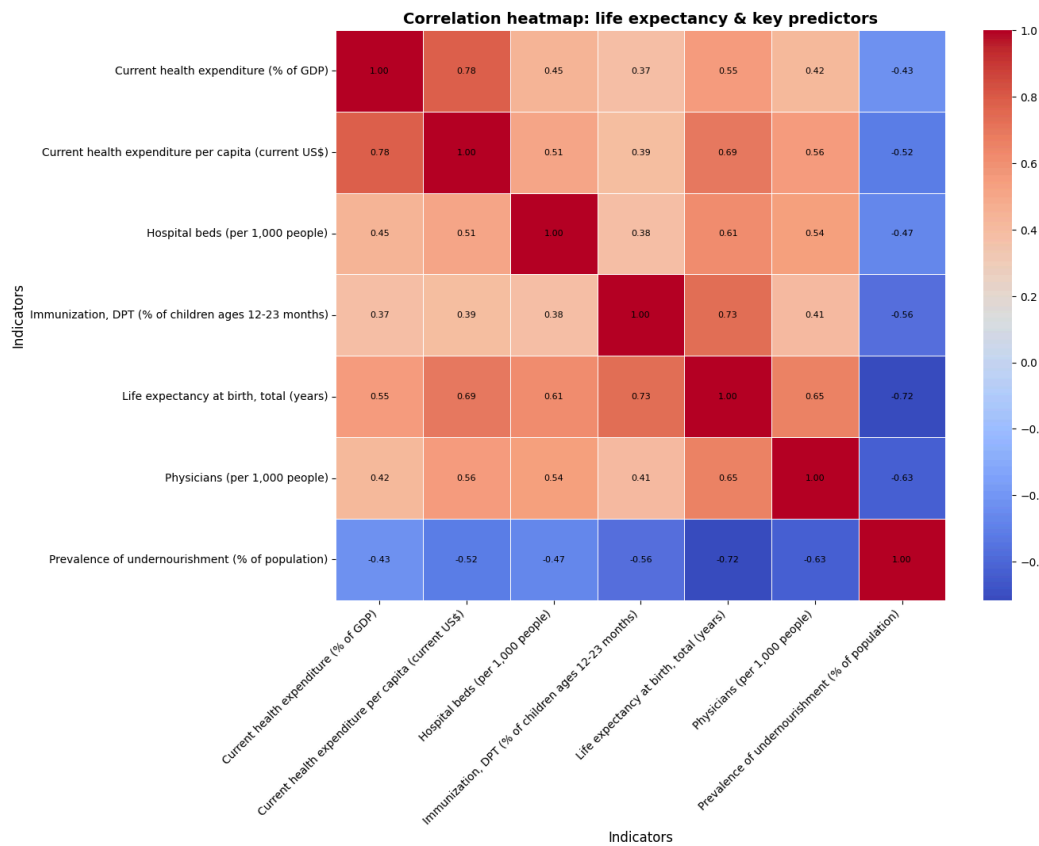


- This graph shows the relationship between life expectancy and healthcare resources (nurses, hospital beds, physicians per 1,000 people). Life expectancy rises rapidly with modest resource increases, but beyond a threshold, the effect diminishes. Outliers show some countries with high resources but lower life expectancy.

9. We explored the relationship between healthcare spending per capita and life expectancy using a scatter plot.

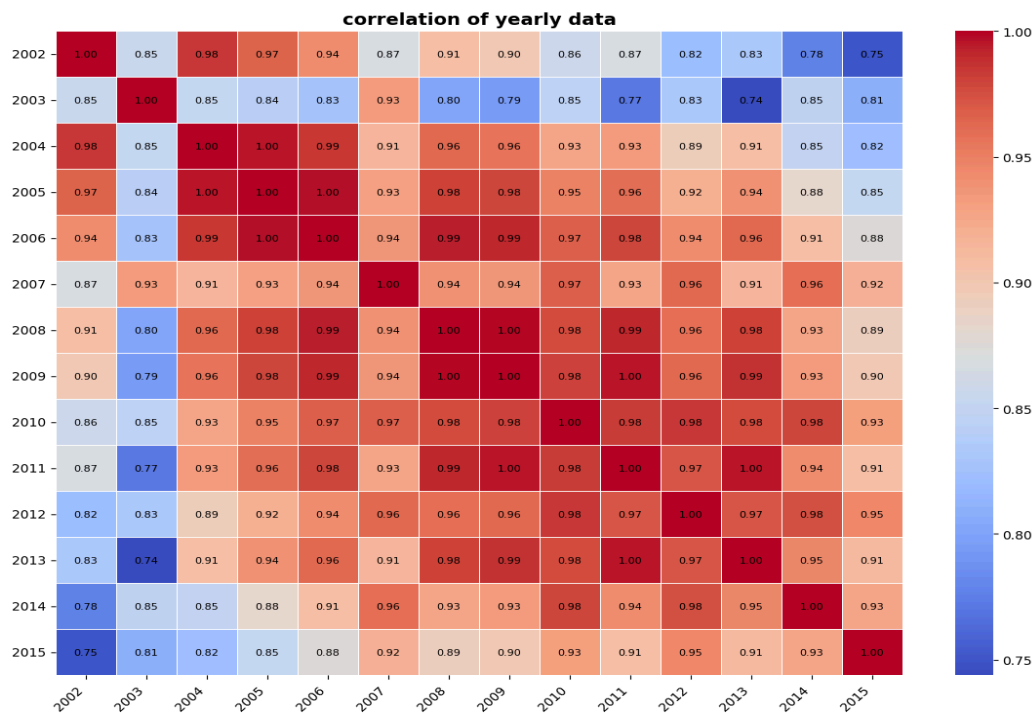


- This graph shows the relationship between life expectancy and healthcare expenditure per capita. Life expectancy rises sharply with lower spending, highlighting the impact of small investments, but plateaus around \$1,000 per capita, indicating diminishing returns. High-income countries cluster at the top with high spending and longevity, while low-income countries show lower spending, shorter life expectancy, and greater variability. Some outliers have high spending but relatively low life expectancy.
10. We created a correlation heatmap to analyze how different healthcare indicators relate to life expectancy.



- Immunization, healthcare expenditure per capita, and physicians per 1,000 people have the strongest positive correlations with life expectancy, meaning that access to medical professionals and preventative care can significantly improve longevity.
- Undernourishment has the strongest negative impact on life expectancy, which indicates that food insecurity directly limits health outcomes.
- Healthcare spending shows diminishing returns. Higher spending is linked to longer life expectancy, but the effect plateaus, as seen with healthcare expenditure as a percentage of GDP.
- Resource availability also correlates positively with life expectancy, meaning that greater access to healthcare infrastructure contributes to longer lifespans.

11. We analyzed how life expectancy trends correlate across different years to identify long-term patterns and stability.



- Life expectancy values from adjacent years are highly correlated, showing consistency in trends over time.
- As the gap between years increases, correlation slightly decreases. This suggests that while life expectancy remains relatively stable over short periods long-term changes may introduce more variability,
- The earliest and latest years have lower correlations with other years, which may indicate shifts in global health trends affecting life expectancy over time.

### Opportunities and challenges in solving the problem statement

- Our analysis shows that small increases in healthcare spending can significantly improve life expectancy in low-income countries. Therefore, policies should focus on cost-effective healthcare solutions, such as preventive care, vaccination programs, and access to basic healthcare facilities to maximize impact with limited resources.
- Instead of simply increasing overall healthcare spending, governments should focus on strategic investments, such as investing in maternal care, sanitation, and early disease detection, which show stronger correlations with life expectancy improvements than just expanding hospital infrastructure.
- The fact that life expectancy increased after the 2008 financial crisis suggests that well-planned healthcare investments and aid programs can protect public health even during financial downturns. Therefore, policymakers should prioritize resilient and long-term healthcare funding models to ensure continued progress regardless of economic fluctuations.



- Despite overall global improvements, low-income countries still face significant gaps in life expectancy. Our heatmap analysis shows that simply increasing the number of doctors and hospital beds is not enough. Governments must prioritize public health policies, disease prevention, and access to clean water to drive long-term improvements.
- The spike-dip pattern in middle-income countries shows that economic instability can disrupt healthcare progress, which makes it harder for these countries to sustain long-term health improvements.
- The spike-dip pattern in middle-income countries shows that economic instability can disrupt healthcare progress, which makes it harder for these countries to sustain long-term improvements.
- Our EDA confirms that life expectancy is influenced by more than just healthcare factors. Financial investment is crucial, but preventive care, sanitation, and economic stability are equally important in increasing longevity. Addressing global healthcare disparities requires a multifaceted approach that combines policy interventions, economic resilience, and public health investments for sustainable improvements in life expectancy worldwide.

### **Data problems identified & recommendations**

Our EDA showed several data challenges that need to be addressed to ensure accurate analysis. One major issue was missing data for certain healthcare indicators, which were not consistently reported across all countries and years. To address this, we handled missing values before splitting the dataset by removing columns with excessive missing data and applying appropriate imputation techniques. For example, we did mode for categorical variables, median for skewed data, and mean for continuous numerical features. This made our dataset remain complete and reliable for analysis.

Additionally, we identified extreme outliers in healthcare expenditure, with some countries spending over \$100,000 per capita, and others spend close to zero. These extreme values could distort our modeling and lead to skewed predictions. To mitigate this, we will apply log transformations to normalize distributions and prevent large disparities. We will also review and cap extreme values where necessary for a more balanced dataset.

Another concern is unequal data distribution across income groups. High-income countries tend to have more complete datasets, and low-income countries often have gaps and inconsistencies, which could introduce bias in our models. To address this, we may use weighted modeling techniques and imputation methods to make sure that our predictions remain reliable for lower-income countries.

Moving forward, our focus will be on using these insights to develop predictive models that can help better understand and address global healthcare inequities.